



Mansoura University
Faculty of Computers and Information
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MODELLING & SIMULATION

SW,BI,AI

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COURSE SYLLABUS

- **Chapter 1: Introduction.**
- **Chapter 2: Probability as Using in Simulation.**
- **Chapter 3: Queueing Simulation.**
- **Chapter 4: Inventory Simulation.**
- **Chapter 5: Random-Number Generation.**
- **Chapter 6: Input Modeling.**
- **Chapter 7: Output Analysis.**

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SYSTEMS

- **What a "system" means:**
- **From Greek "systema"** - organized relationship among functioning units
- Any set of interrelated components acting together for a common objective
- Group of objects joined in regular interaction toward a purpose
- A system is not just a collection of parts - the parts must interact and work toward a common goal.

•**System:** A car (engine, wheels, transmission work together for transportation)

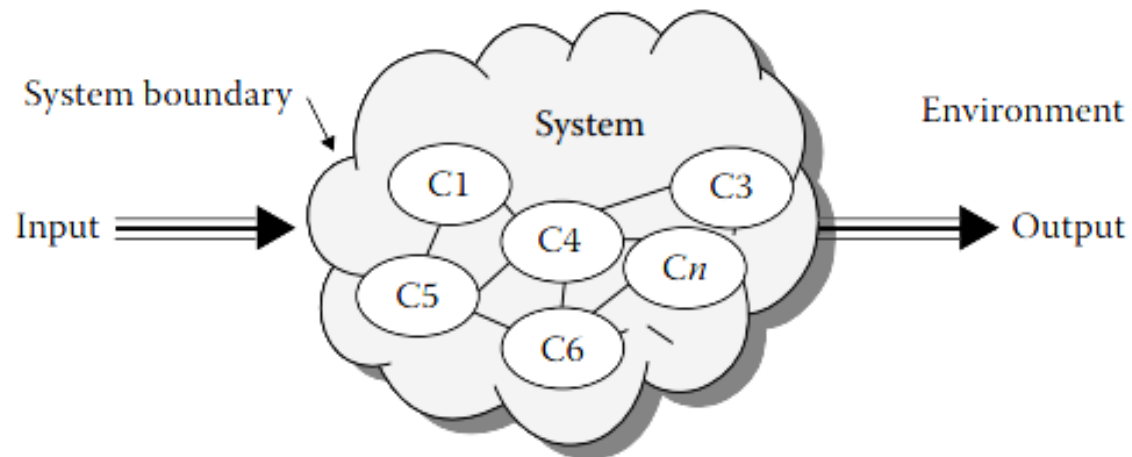
•**NOT a system:** Random people on the street (no common objective)

SYSTEMS

- The collection of entities that compose a system for one study might be **only a subset of another larger system.**
- System definition is not absolute - it's determined by what questions you're trying to answer.
 - **For example,**
 - **Study 1:** "How many tellers do we need?"
 - System = tellers + customers in teller area
 - **Study 2:** "How does loan processing affect overall bank performance?"
 - System = tellers + customers + loan officers + safety deposit

SYSTEM ENVIRONMENT

- **A system is characterized by the following attributes:**
 - ✓ **System boundary** - separates system from environment
 - ✓ **System components** - parts that make up the system
 - ✓ **Environment** - everything outside the boundary that affects the system

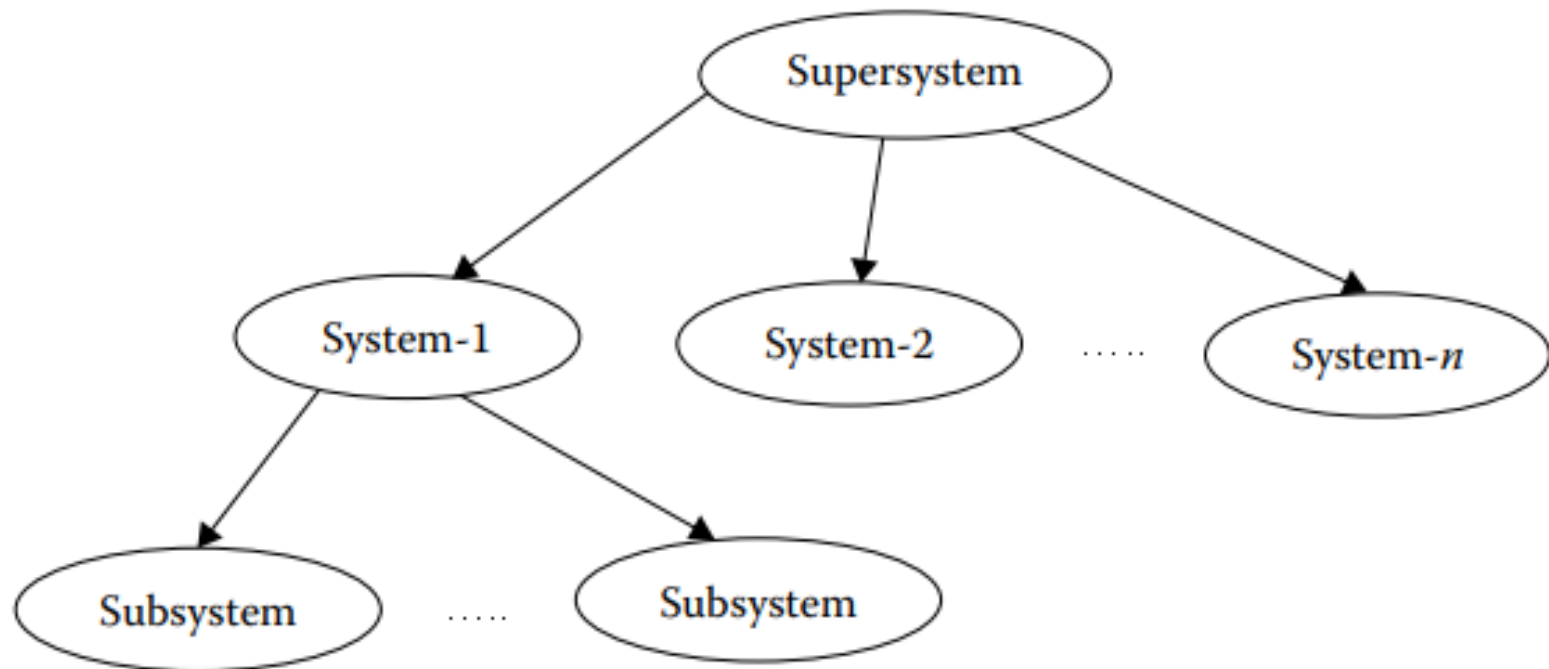


SYSTEM ENVIRONMENT

- A system is often affected by **changes occurring outside the system**. Such changes are said to occur in the system environment.
 - ✓ Factory :Arrival orders
 - ✓ Banks :Arrival of customers

SYSTEM ENVIRONMENT

- Every system consists of **subsystem** or components at lower levels and **supersystems** at higher levels.



COMPONENTS OF A SYSTEM

- An **entity** is an object of interest in the system.
 - ✓ Machines in factory.
 - ✓ Customers in Bank.
- An **attribute** is a property of an entity.
 - ✓ Speed, capacity.
 - ✓ Balance in customer accounts.
- An **activity** represents a time period of specified length.
 - ✓ Welding.
 - ✓ Making deposits.

Hospital:

- **Entities:** Patients, doctors, nurses, beds
- **Attributes:** Patient's age/condition, doctor's specialty, bed availability
- **Activities:** Examination (15 min), surgery (2 hours), recovery (3 days)

COMPONENTS OF A SYSTEM

- The **state** of a system is defined to be that collection of variables necessary to describe the system at any time, relative to the objectives of the study.
 - ✓ Status of machine (busy, idle, down,...)
 - ✓ In the study of a bank, possible state variables are the number of busy tellers, the number of customers waiting in line or being served, and the arrival time of the next customer.

COMPONENTS OF A SYSTEM

- **Event:** Immediate occurrence that might change system state
 - **Endogenous:** Occurs within system (service completion)
 - **Exogenous:** Occurs in environment (customer arrival)
 - ✓ In the bank study, the arrival of a customer is an exogenous event, and the completion of service of a customer is an endogenous event.

COMPONENTS OF A SYSTEM

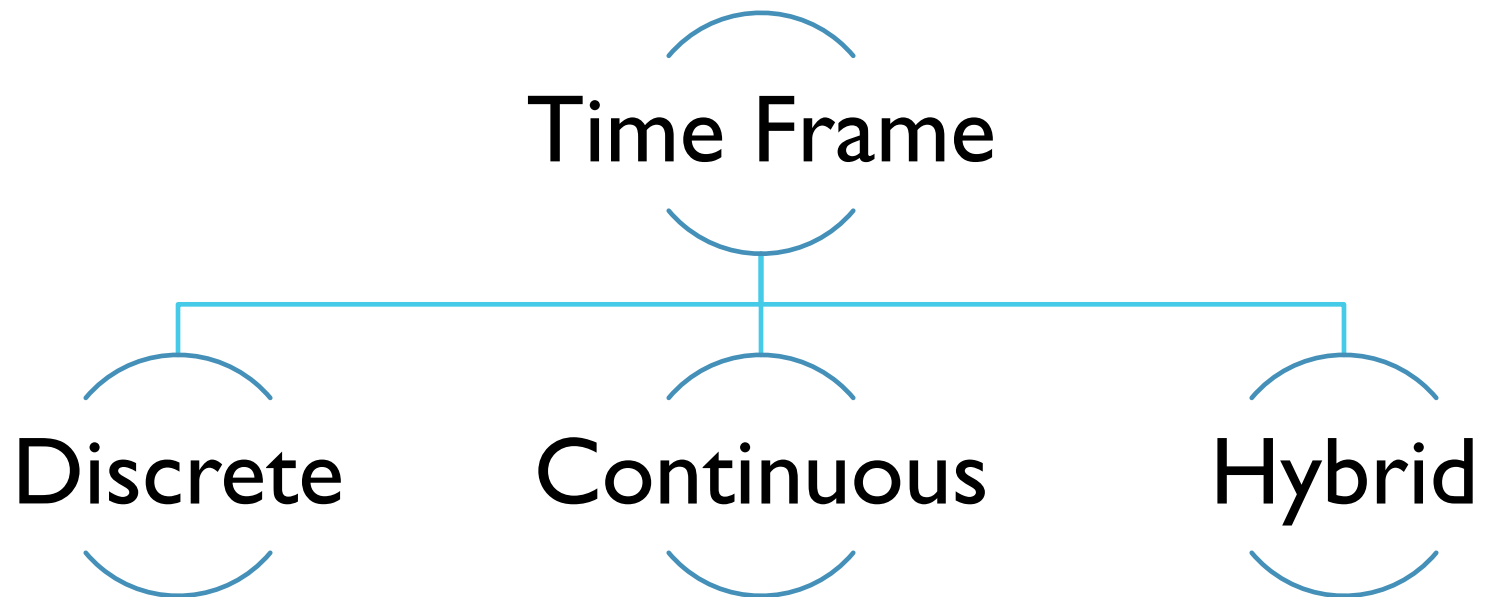
- Examples of entities, attributes, activities, events, and state variables for several systems.

<i>System</i>	<i>Entities</i>	<i>Attributes</i>	<i>Activities</i>	<i>Events</i>	<i>State Variables</i>
Banking	Customers	Checking-account balance	Making deposits	Arrival; departure	Number of busy tellers; number of customers waiting
Production	Machines	Speed; capacity; breakdown rate	Welding; stamping	Breakdown	Status of machines (busy, idle, or down)
Communications	Messages	Length; destination	Transmitting	Arrival at destination	Number waiting to be transmitted
Inventory	Warehouse	Capacity	Withdrawing	Demand	Levels of inventory; backlogged demands

CLASSIFICATION OF SYSTEMS

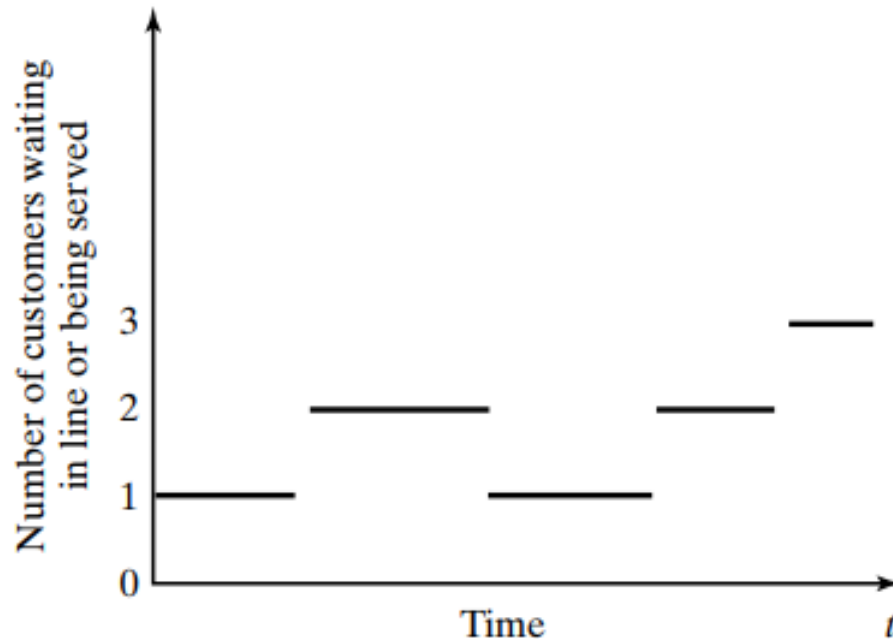
- **Systems can be classified based on time frame, type of measurements taken, type of interactions, nature, type of components, etc.**

CLASSIFICATION OF SYSTEMS



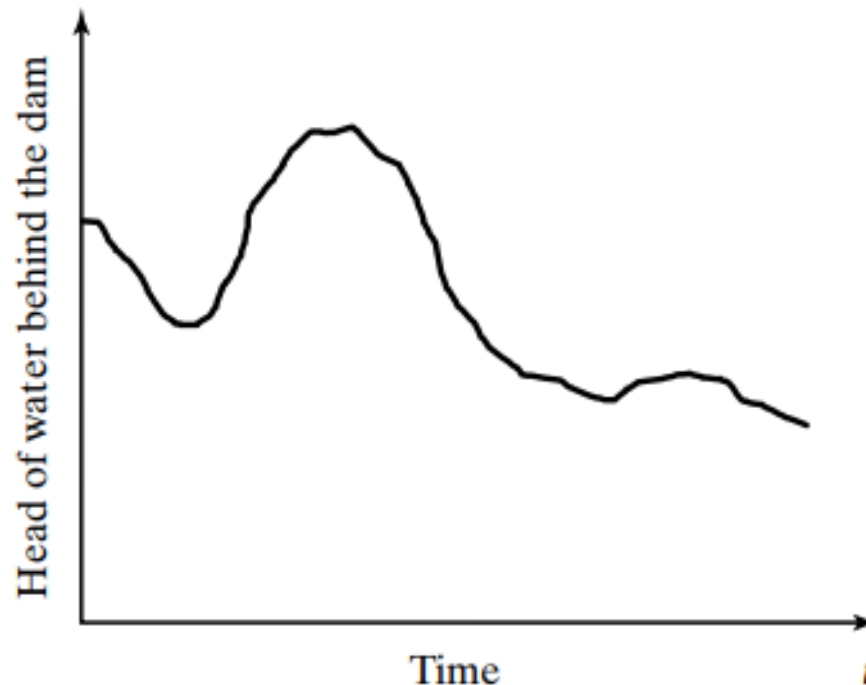
A DISCRETE SYSTEM

- The state variable(s) change only at discrete points in time, for example, queuing systems (bank, telephone network, traffic lights, machine).



A CONTINUOUS SYSTEM:

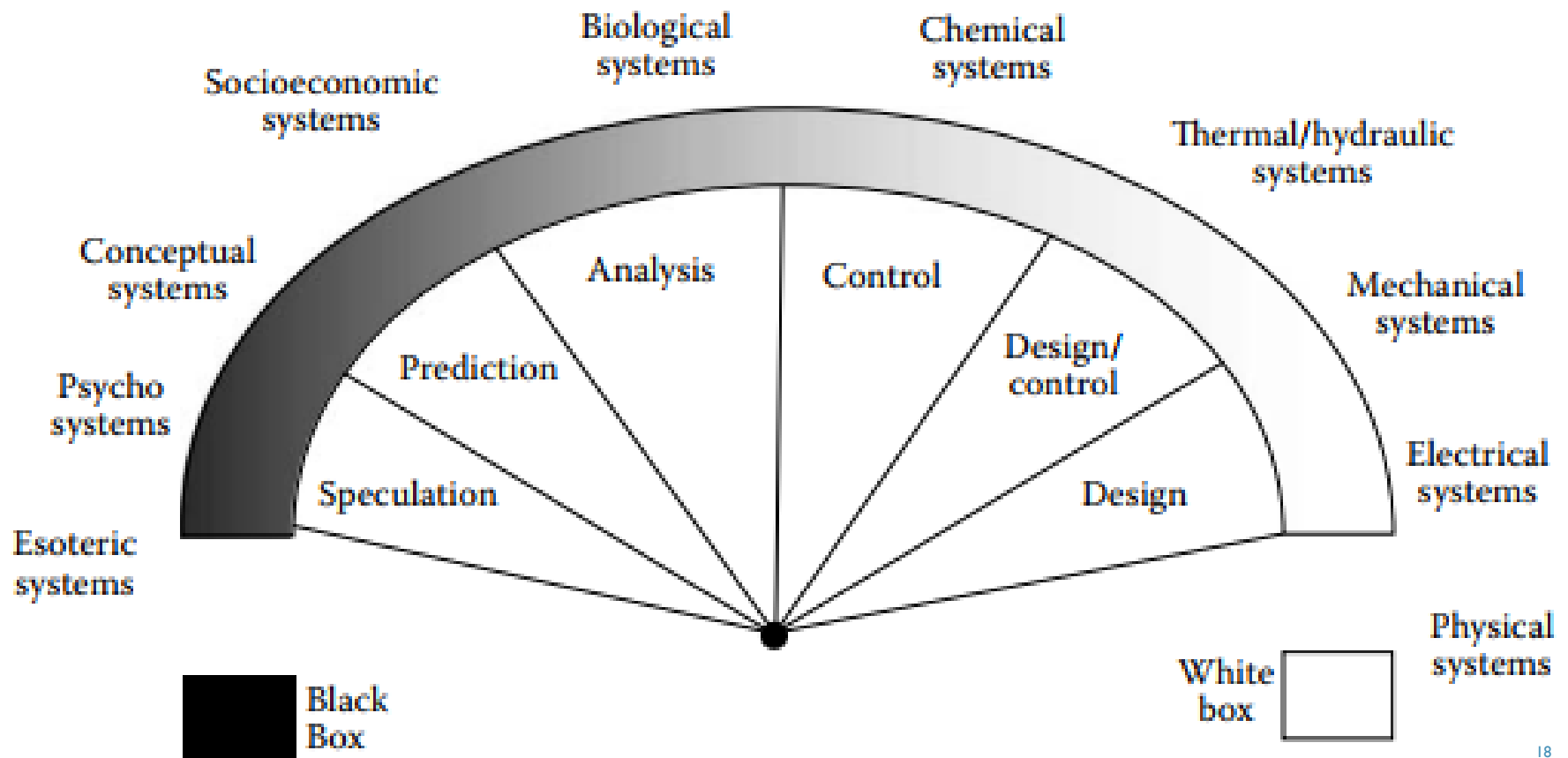
- The state variable(s) change continuously over time. An example is the head of water behind a dam, solar system, charging a battery.



A HYBRID SYSTEM:

- A hybrid system is a combination of continuous and discrete dynamic system behavior.
- A hybrid system has the benefit of encompassing a larger class of systems within its structure, allowing more flexibility in modeling continuous and discrete dynamic phenomena.
- For example, traffic along a road with traffic lights.

CLASSIFICATION ACCORDING TO THE COMPLEXITY OF THE SYSTEM:

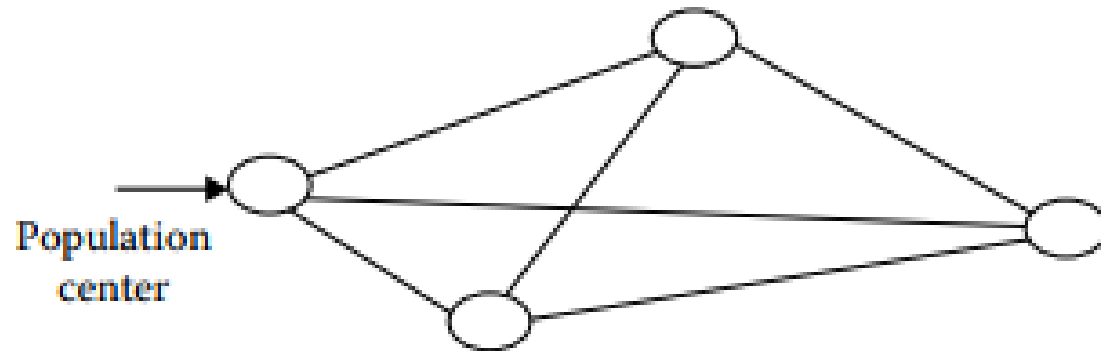


PHYSICAL SYSTEMS:

- Physical systems can be defined as systems whose variables can be measured with physical devices that are quantitative such as electrical systems, mechanical systems, computer systems, thermal systems, or a combination of these systems.
- Physical system is a collection of components, in which each component has its own behavior, used for some purpose.
- These systems are relatively less complex.

CONCEPTUAL SYSTEMS:

- Conceptual systems are those systems in which all the measurements are conceptual or imaginary and in qualitative form as in psychological systems, social systems, health care systems, and economic systems. These are complex systems.



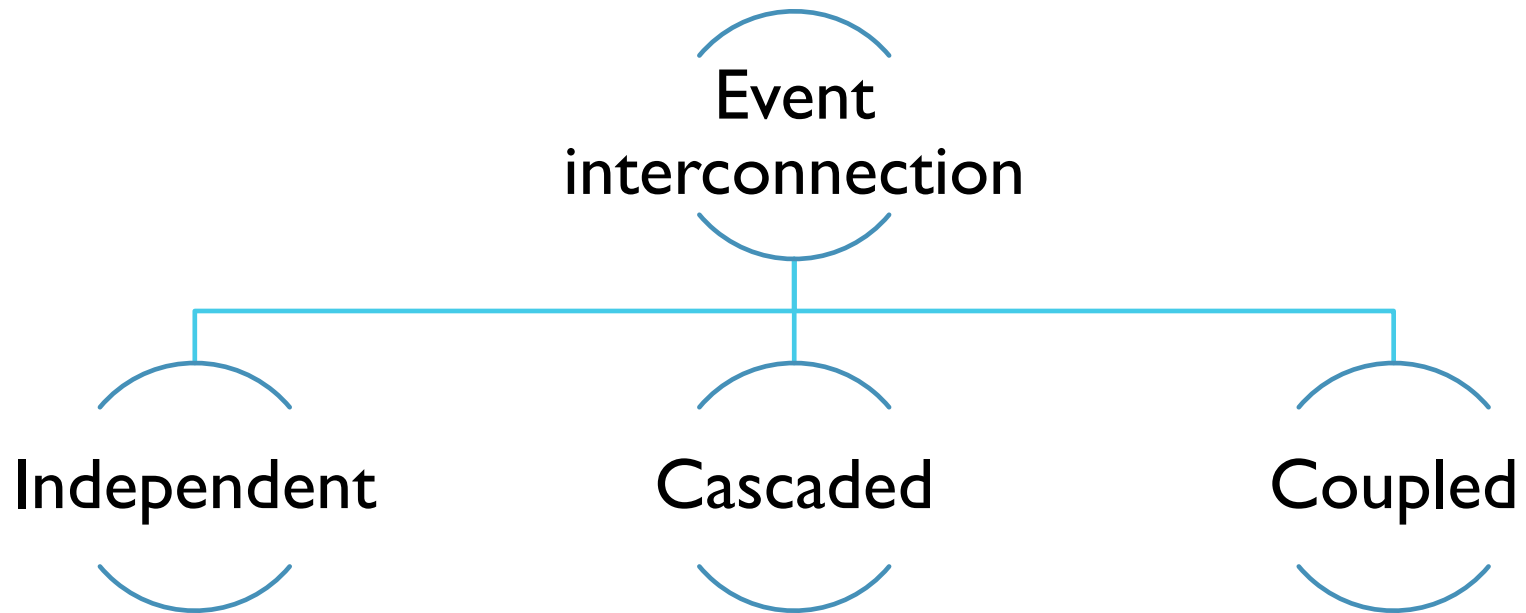
Transportation system

ESOTERIC SYSTEMS:

- Esoteric systems are the systems in which the **measurements are not possible with physical** measuring devices.
- The complexity of these systems is of highest order.

ACCORDING TO THE DEGREE OF INTERCONNECTION OF EVENTS.

- Systems will be divided into three classes according to the degree of interconnection of events.



INDEPENDENT SYSTEM:

- If the events have no effect upon one another,
then the system is classified as independent.

CASCADED SYSTEM:

- If the effects of the events are unilateral (that is, part A affects part B, B affects C, C affects D, and not vice versa), the system is classified as cascaded.

COUPLED SYSTEM:

- If the events mutually affect each other, the system is classified as coupled.

NATURE AND TYPE OF COMPONENTS

- **Systems can be classified according to the Nature and Type of Components.**
 - Static or dynamic components
 - Linear or nonlinear components
 - Deterministic or stochastic components
 - Continuous-time and discrete-time systems
 - Others...

STATIC OR DYNAMIC COMPONENTS

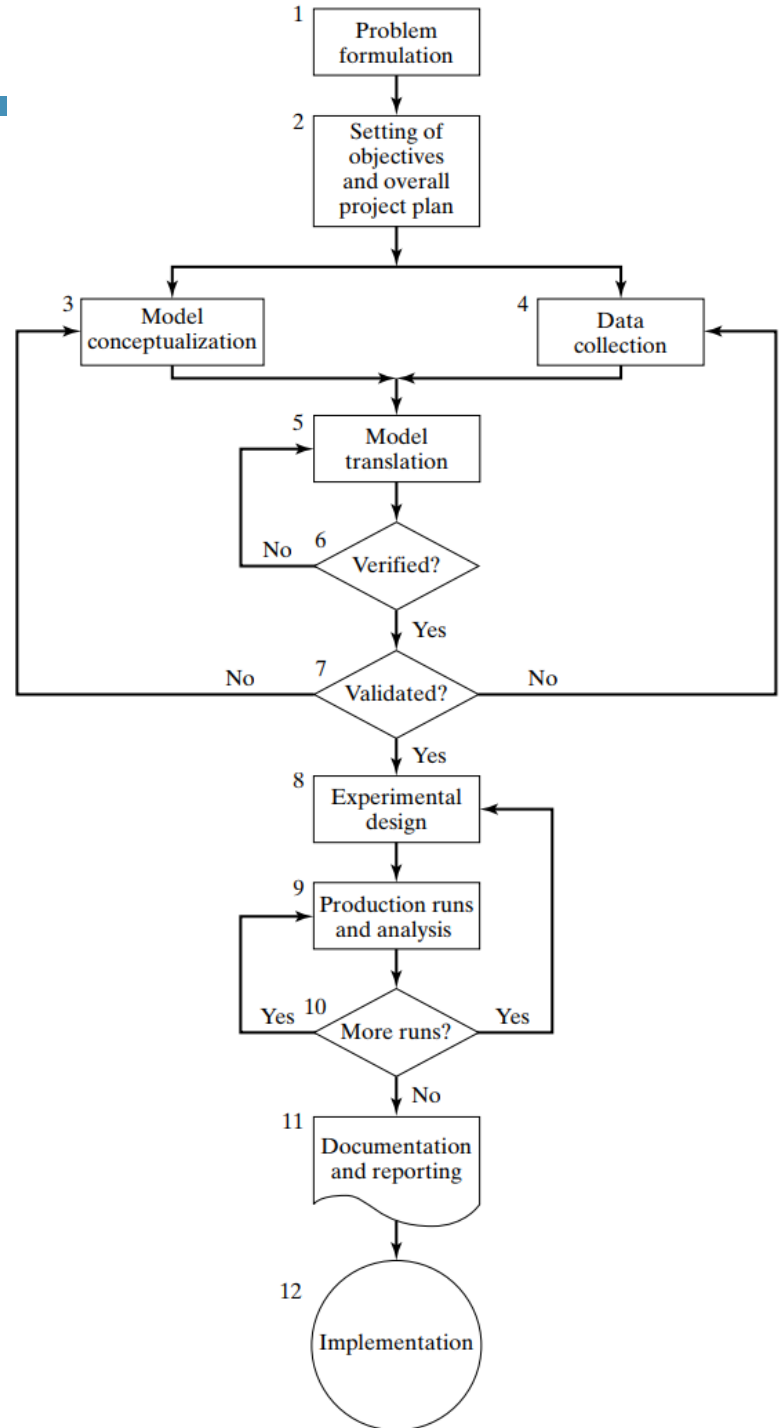
- **A static simulation model**, sometimes called a **Monte Carlo simulation**, represents a system at a particular point in time.
- **Dynamic simulation models** represent systems as they change over time. The simulation of a bank from 9:00A.M. to 4:00P.M. is an example of a dynamic simulation.

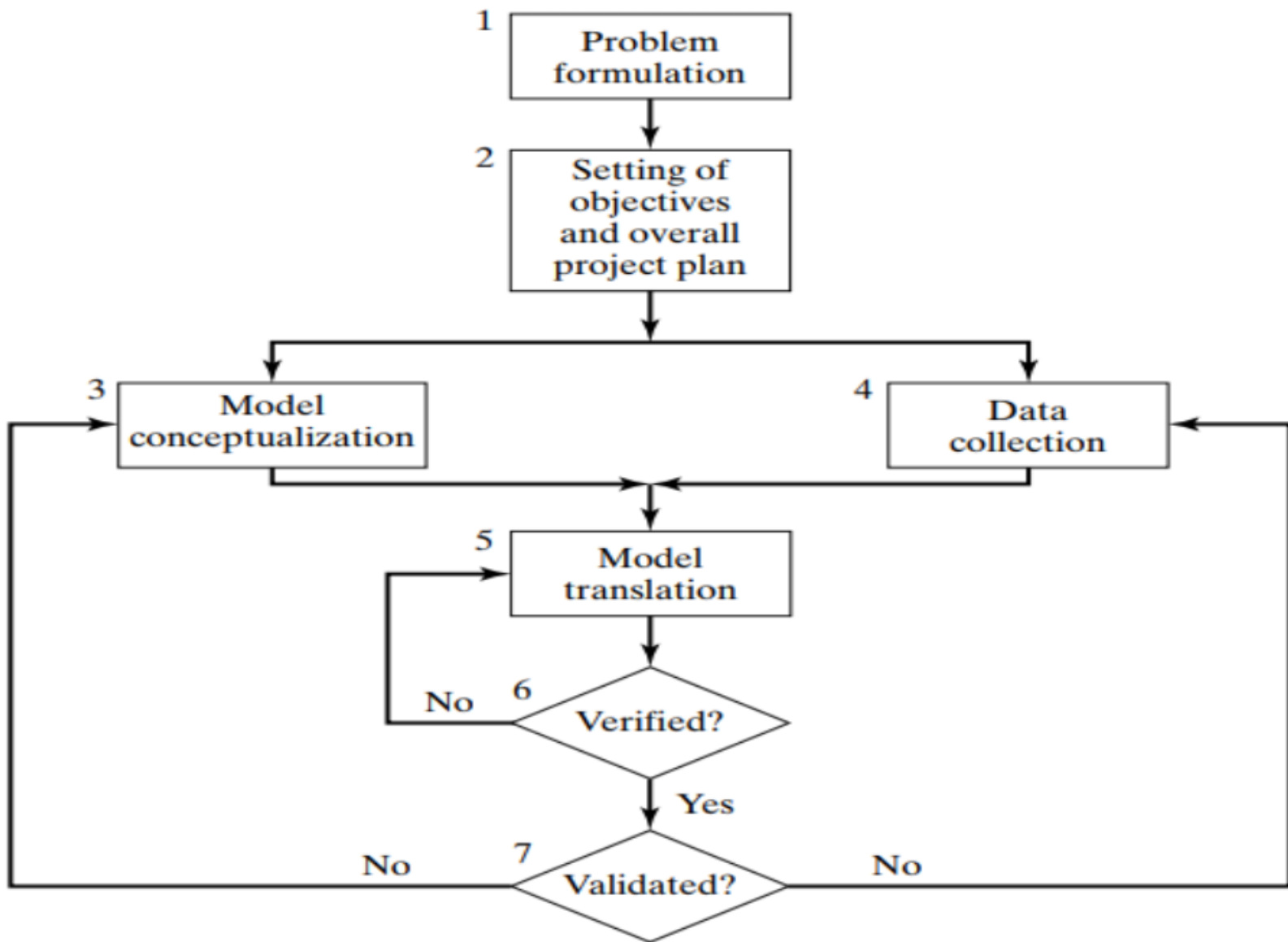
DETERMINISTIC OR STOCHASTIC COMPONENTS

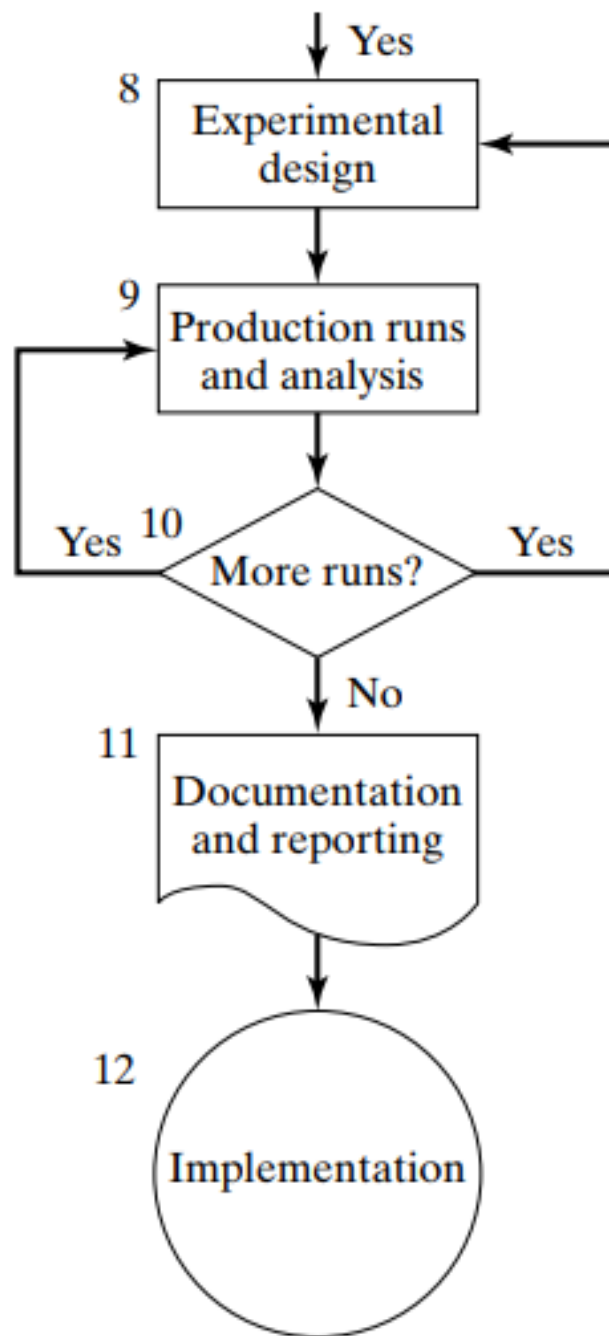
- Simulation models that contain no random variables are classified as **deterministic**. No probabilistic component in the system.
- A **stochastic** simulation model has one or more random variables as inputs. Some components of the system has a probabilistic behavior (Random variable, event probability). Example: Queuing systems.

Steps in a Simulation Study

1. Problem formulation
2. Setting of objectives and overall project plan
3. Model conceptualization
4. Data collection
5. Model translation
6. Verified?
7. Validated?
8. Experimental design
9. Production runs and analysis
10. More runs?
11. Documentation and reporting
12. Implementation







I. PROBLEM FORMULATION

- Every study should begin with a **statement of the problem**.
- If the statement is provided by the **policymakers** or those that have the problem, the analyst must ensure that the problem being described is clearly understood.
- If a problem statement is being developed by the **analyst**, it is important that the policymakers understand and agree with the formulation.

2. SETTING OF OBJECTIVES AND OVERALL PROJECT PLAN

- The objectives indicate the **questions to be answered by simulation**.
- At this point, a **determination should be made concerning whether simulation is the appropriate methodology** for the problem as formulated and the objectives as stated.

3. MODEL CONCEPTUALIZATION

- It is best to start with a **simple model** and build toward **greater complexity**. However, the model complexity need not exceed that required to accomplish the purposes for which the model is intended.
- It is advisable to involve the model user in model conceptualization. Involving the model user will both enhance the quality of the resulting model and increase the confidence of the model user in the application of the model.

4. DATA COLLECTION

- There is a constant interplay between the construction of the model and the collection of the needed input data.
- As the complexity of the model changes, the required data elements can also change.
- Also, since data collection takes such a large portion of the total time required to perform a simulation, it is necessary to begin as early as possible, usually together with the early stages of model building.

5. MODEL TRANSLATION

- Most real-world systems result in models that require a great deal of information storage and computation, **so the model must be entered into a computer-recognizable format.**
- We use the term program even though it is possible, in many instances, to accomplish the desired result with little or no actual coding.
- The modeler must decide whether to program the model in a simulation language or to use special-purpose simulation software.

6. VERIFIED?

- Did we build the model right?
- Verification pertains to the computer program that has been prepared for the simulation model.
- Is the computer program performing properly? With complex models, it is difficult, if not impossible, to translate a model successfully in its entirety without a good deal of debugging; if the input parameters and logical structure of the model are correctly represented in the computer, verification has been completed.

7.VALIDATED?

- Did we build the right model?
- Validation usually is achieved through the calibration of the model, an iterative process of comparing the model against actual system behavior and using the conflict between the two, and the insights gained, to improve the model.This process is repeated until model accuracy is judged acceptable.

8. EXPERIMENTAL DESIGN

- The alternatives that are to be simulated must be determined.
- Often, the decision concerning which alternatives to simulate will be a function of runs that have been completed and analyzed.
- For each system design that is simulated, decisions need to be made concerning the length of the **initialization period, the length of simulation runs, and the number of replications to be made of each run.**

9. PRODUCTION RUNS AND ANALYSIS

- Production runs and their subsequent analysis, are used to estimate measures of performance for the system designs that are being simulated.

10. MORE RUNS?

- Given the analysis of runs that have been completed, the analyst determines whether additional runs are needed and what design those additional experiments should follow.

II. DOCUMENTATION AND REPORTING

- **There are two types of documentation: program and progress.**
- Program documentation is necessary for numerous reasons.
 - If the program is going to be used again by the same or different analysts, it could be necessary to understand how the program operates.
 - Also, if the program is to be modified by the same or a different analyst.

II. DOCUMENTATION AND REPORTING

- **Progress reports** provide the important, written history of a simulation project.

II. DOCUMENTATION AND REPORTING

- “it is better to work with many intermediate milestones than with one absolute deadline.”
- Possibilities prior to the final report include a model specification, prototype demonstrations, animations, training results, intermediate analyses, program documentation, progress reports, and presentations.

12. IMPLEMENTATION

- The success of the implementation phase depends on how well the previous eleven steps have been performed.

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CHAPTER 2 CONTENT

- Introduction.
- Discrete random variables.
 - Probability mass function.
 - Mean, Variance, and Standard deviation.
- Continuous random variables.
 - Probability density function.
 - Expectation.
- Cumulative distribution function.

RANDOM (STATISTICAL) EXPERIMENT:

- An experiment <with known outcomes> whose outcome cannot be predicted with certainty, before the experiment is run.



SAMPLE SPACE (S):

- Set of **ALL** possible outcomes of a random experiment.
- A sample space is **discrete** if it consists of a finite or countable infinite set of outcomes.
- A sample space is **continuous** if it contains an interval (either finite or infinite) of real numbers.

SAMPLE SPACE (S):

Sample Space (S):

Discrete

- Set of **ALL** possible outcomes of a random experiment.



The roll of a dice

$$S = \{1, 2, 3, 4, 5, 6\}$$

Each outcome in a sample space is called an **element** or a **member** of the sample space, or simply a **sample point**.

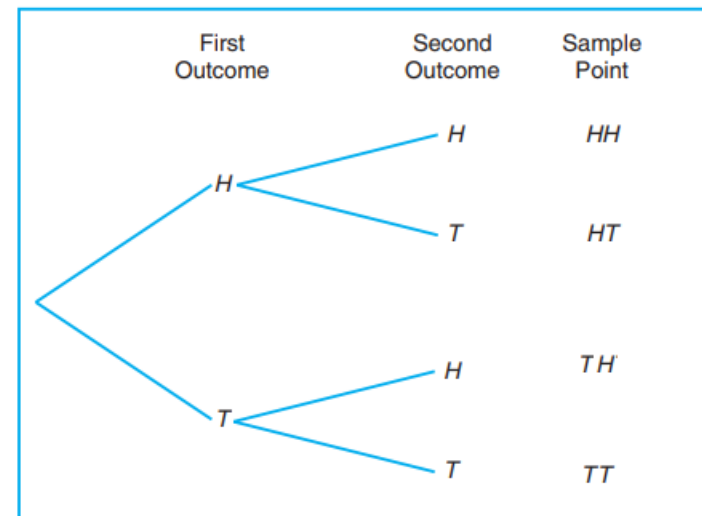
SAMPLE SPACE (S):

Example1:

Find the sample space for the random experiments (flipping) a coin of two times?

Answer:

$$S = \{HH, HT, TH, TT\}$$



SAMPLE SPACE (S):

Example3:

Continuous

Consider an experiment that selects a cell phone camera and records the recycle time of a flash (the time taken to ready the camera for another flash).

$$S = R^+ = \{x \mid x > 0\}$$

If it is known that all recycle times are between 1.5 and 5 seconds, the sample space can be

$$S = \{x \mid 1.5 < x < 5\}$$

EVENT (E):

- A result of *none* , *one* , or *more outcomes in the sample* space. An event is a subset of the sample space of a random experiment.



The roll of a dice

$$S = \{1,2,3,4,5,6\}$$
$$E = \{2,4,6\}$$

Even Numbers

PROBABILITY

- **Probability is used to quantify the likelihood, or chance**, that an outcome of a random experiment will occur. “The chance of rain today is 30%” is a statement that quantifies our feeling about the possibility of rain.
- The likelihood of an outcome is quantified by assigning a number from the **interval $[0, 1]$** to the outcome (or a percentage from 0 to 100%).
- Higher numbers indicate that the outcome is more likely than lower numbers.
- A 0 indicates an outcome will not occur. A probability of 1 indicates that an outcome will occur with certainty.

EQUALLY LIKELY OUTCOMES:

- Whenever a sample space consists of N possible outcomes that are equally likely, the probability of each outcome is $1/N$.

$$\text{Ex. } S = \{1, 2, 3, 4, 5, 6\}$$

$$P(1) = \frac{1}{6}, \quad P(2) = \frac{1}{6}, \quad \dots, \quad P(6) = \frac{1}{6}$$

EQUALLY LIKELY OUTCOMES:

Equally Likely Outcomes:

- S Is a sample space, E is an event
- $E \subseteq S$

$$P(E) = \frac{\text{\#of outcomes in event}(E)}{\text{Total \# of outcomes in sample space}(S)} = \frac{n(E)}{n(S)}$$

$$1 \geq P(E) \geq 0$$

CONDITIONAL PROBABILITY

- Sometimes probabilities need to be reevaluated as additional information becomes available.
- The probability of an event B under the knowledge that the outcome will be in event A is denoted as

$$P(B | A)$$

and this is called the conditional probability of B given A .

$$P(B|A) = \frac{P(A \cap B)}{P(A)} , \quad \text{provided } P(A) > 0$$

Equally Likely Outcomes

$$\frac{P(A \cap B)}{P(A)} = \frac{\text{number of outcomes in } A \cap B}{\text{number of outcomes in } A}$$

DISJOINT (OR MUTUALLY EXCLUSIVE):

S is a sample space, A, B are two events

$A, B \subseteq S$ and A, B **Disjoint or mutually exclusive**

$$\therefore P(A \cap B) = 0$$

$$\therefore P(A \cup B) = P(A) + P(B)$$

$$\therefore P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0}{P(B)} = 0$$

INDEPENDENCE:

S is a sample space, A, B are two events
 $A, B \subseteq S$ and A, B are **independent**

$$\therefore P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) * P(B)}{P(B)} = P(A)$$

$$\therefore P(A \cap B) = P(A) * P(B)$$

$$\therefore P(A \cup B) = P(A) + P(B) - P(A) * P(B)$$

RANDOM VARIABLE

- **Random Variable**
- Is a function that assigns a real number to each outcome in the sample space of random experiment. Denoted by an uppercase letter such as X
- **A Discrete Random Variable**
- Is a random variable with a finite (or countable infinite) range.
- The possible values of X may be listed as $x_1, x_2, \dots$

RANDOM VARIABLE

Example1

- Flipping a coin of two times. Let X is the number of heads.

Answer:

$$S = \{HH, HT, TH, TT\}$$
$$\quad \quad \quad 2 \quad 1 \quad 1 \quad 0$$

$$x = 0, 1, 2$$

$$P(0) = \frac{1}{4}, \quad P(1) = \frac{2}{4}, \quad P(2) = \frac{1}{4}$$

PROBABILITY MASS FUNCTION

For a discrete random variable X with possible values $x_1, x_2, \dots, x_n$, a **probability mass function** is a function such that

$$(1) \quad f(x_i) \geq 0$$

$$(2) \quad \sum_{i=1}^n f(x_i) = 1$$

$$(3) \quad f(x_i) = P(X = x_i)$$

CUMULATIVE DISTRIBUTION

- The cumulative distribution function (cdf), denoted by $F(x)$, measures the probability that the random variable X assumes a value less than or equal to x , that is,

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

If X is discrete, then

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

x	-2	-1	0	1	2
$f(x) = P(X = x)$	1/8	2/8	2/8	2/8	1/8
$F(x) = P(X \leq x)$	1/8	3/8	5/8	7/8	8/8

MEAN AND VARIANCE

- Two numbers are often used to summarize a probability distribution for a random variable X .
- The **mean** is a measure of the center or middle of the probability distribution,
- and the **variance** is a measure of the dispersion, or variability in the distribution.

The **mean** or **expected value** of the discrete random variable X , denoted as μ or $E(X)$, is

$$\mu = E(X) = \sum_x xf(x)$$

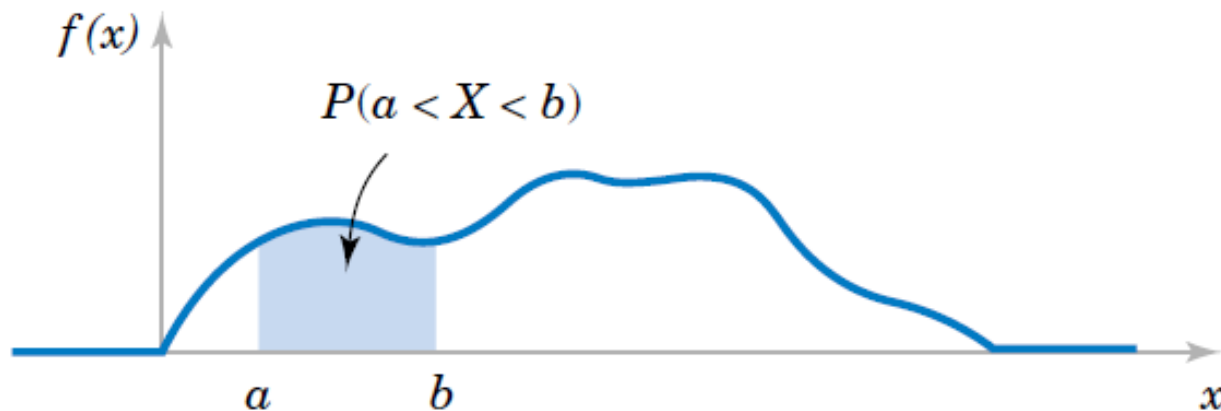
The **variance** of X , denoted as σ^2 or $V(X)$, is

$$\sigma^2 = V(X) = E(X - \mu)^2 = \sum_x (x - \mu)^2 f(x) = \sum_x x^2 f(x) - \mu^2$$

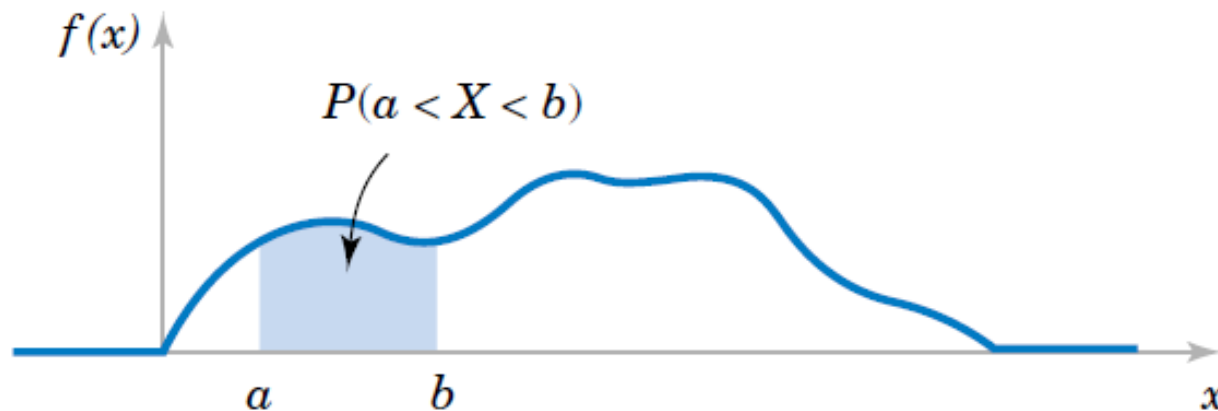
The **standard deviation** of X is $\sigma = \sqrt{\sigma^2}$.

CONTINUOUS RANDOM VARIABLE:

- If the range space R_X of the random variable X is an **interval or a collection of intervals**, X is called a *continuous random variable*.
- A continuous random variable has a probability of **0** of assuming *exactly any of its values*. Consequently, its probability distribution cannot be given in tabular form.



CONTINUOUS RANDOM VARIABLE:



If X is a **continuous random variable**, for any x_1 and x_2 ,

$$P(x_1 \leq X \leq x_2) = P(x_1 < X \leq x_2) = P(x_1 \leq X < x_2) = P(x_1 < X < x_2)$$

PROBABILITY DENSITY FUNCTION

For a continuous random variable X , a **probability density function** is a function such that

$$(1) \quad f(x) \geq 0$$

$$(2) \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

$$(3) \quad P(a \leq X \leq b) = \int_a^b f(x) dx = \text{area under } f(x) \text{ from } a \text{ to } b$$

for any a and b

Cumulative Distribution Function

The **cumulative distribution function** of a continuous random variable X is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du$$

for $-\infty < x < \infty$.



Thanks
Any questions